

e-CRF Specifications, Annotated CRF, Test Script Portfolio Project

Project Overview

End-to-End Clinical Data Management Workflow

Following protocol approval in a clinical trial, the Clinical Data Management (CDM) team initiates the study setup phase by drafting the Data Management Plan (DMP). Guided by the protocol and DMP requirements, data management professionals build the core infrastructure of the electronic Case Report Form (eCRF).

To demonstrate practical proficiency in these foundational CDM deliverables, I completed a self-initiated project detailing the complete eCRF design and validation workflow:

- CRF Specifications & Annotations: Authored detailed CRF specifications outlining required data fields, formats, and entry guidelines. Created annotated CRFs serving as a blueprint for database programmers to map data fields accurately to standard data models (such as CDISC SDTM).

- Edit Check Specifications: Drafted comprehensive edit check specifications to ensure data integrity, specifying logical rules, query messaging

-UAT Test Scripting: Developed User Acceptance Testing (UAT) test scripts with predefined steps, inputs, and expected outcomes to validate that programmed edit checks execute correctly prior to study go-live.

In this project, I have drafted CRF specifications, annotations, edit check specifications and test scripts for Demography form, Routine Haematology form and Routine Clinical Chemistry form.

As an aspiring Clinical Data Associate/Coordinator, mastering these core methodologies demonstrates hands-on familiarity with the essential clinical trial lifecycle, data quality controls, and interdisciplinary collaboration required in entry-level CDM roles.

BLUE HIGHLIGHTED TEXT: ANNOTATIONS

YELLOW HIGHLIGHTED TEXT: REFERENCE RANGE

RED UNDERLINED TEXT: TEST SCRIPT

GREY HIGHLIGHTED TEXT : SPECIFICATION

FOLDER: SCREENING VISIT

FORM: DEMOGRAPHY [DM]

GENDER [DM_GD] : MALE - 0
FEMALE -1

PROVIDE CHECKBOX FOR THE OPTIONS

IF none of the checkboxes selected
RAISE QUERY

DATE OF BIRTH : ___/___/___
[DM_DOB]

Entry Should be in DD/MM/YY

[6 digit]format,

Allow only numerical values.

IF ALPHABET /DECIMAL ENTRY
MADE,RAISE QUERY.

IF NO ENTRY MADE,RAISE QUERY.

RACE [DM_RC] :

CAUCASIAN -0
BLACK-1
ORIENTAL-2
OTHERS-3

PROVIDE CHECKBOX FOR THE
OPTIONS

IF NONE OF THE CHECKBOXES
SELECTED,RAISE QUERY

ETHNIC ORIGIN [DM_EO] :

ASIAN -0
BLACK-1
WHITE-2
INDIGENOUS-3
HISPANIC-4

PROVIDE CHECKBOX FOR THE
OPTIONS

IF NONE OF THE CHECKBOXES
SELECTED WITH NO SPECIFIC
REASON THEN RAISE QUERY

IF NONE OF THE CHECKBOXES
SELECTED, and If ethnic origin
can't be disclosed,entry can be
left blank

CRF ANNOTATION FOR DEMOGRAPHY FORM [DM]

FIELD NAME	FIELD LABEL
DM_GD	GENDER
DM_DOB	DATE OF BIRTH
DM_RC	RACE
DM_EO	ETHNIC ORGIN

ROUTINE HAEMATOLOGY[RH] FORM

HEMATOCRIT: __%

[RH_HCT]

FOR MALES 41-50

FEMALES 36 – 44

ENTRY SHOULD BE A NUMERICAL VALUE ,LIMITED TO 3 DIGITS,DECIMALS CAN BE ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF LESS THAN 2 DIGITS OR MORE THAN 3 DIGITS ENTERED

GO TO screening visit folder,go to DEMOGRAPHY FORM ,IN GENDER i.e DM GD, CHECK option chosen:
IF MALE - RAISE QUERY IF VALUE ENTERED 40 .
IF FEMALE- RAISE QUERY IF VALUE ENTERED 35

HAEMOGLOBIN

(RH_HB): __g/dL

FOR MALES 13.5-16,

FEMALES 12.5-14.5

ENTRY SHOULD BE NUMERICAL VALUE,LIMITED TO 3 DIGITS. DECIMALS CAN BE ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF LESS THAN 2 DIGITS OR MORE THAN 3 DIGITS ENTERED

RAISE QUERY IF VALUE ENTERED 13.5 AND BELOW/16 AND ABOVE

GO TO screening visit folder, go to DEMOGRAPHY FORM ,IN GENDER i.e DM GD, CHECK option chosen:
IF MALE - RAISE QUERY IF VALUE ENTERED 13.4.
IF FEMALE- RAISE QUERY IF VALUE ENTERED 12.4

MEAN CELLULAR HAEMOGLOBIN

(RH_MCH): __pg

FOR BOTH MALES AND FEMALES , 27-32

ENTRY SHOULD BE NUMERICAL VALUES,LIMITED TO 3 DIGITS.DECIMALS ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF LESS THAN 2 DIGITS OR MORE THAN 3 DIGITS ENTERED

RAISE QUERY IF VALUE ENTERED 26.

MEAN CELLULAR HAEMOGLOBIN CONCENTRATION

(RH_MCHC):__g/dL

FOR BOTH MALES AND FEMALES , 31.5-34.5

ENTRY SHOULD BE NUMERICAL
VALUES,LIMITED TO 3 DIGITS.DECIMALS
ALLOWED. RAISE QUERY IF ALPHABET
INPUT ENTERED.

RAISE QUERY IF LESS THAN 2 DIGITS OR
MORE THAN 3 DIGITS ENTERED

entered data 31 then RAISE QUERY

MEAN CELLULAR VOLUME (RH_MCV): __ fL

FOR BOTH MALES AND FEMALES , 83-101,

ENTRY SHOULD BE NUMERICAL
VALUES,LIMITED TO 3 DIGITS.DECIMALS
ALLOWED. RAISE QUERY IF ALPHABET
INPUT ENTERED.

RAISE QUERY IF LESS THAN 2 DIGITS OR
MORE THAN 3 DIGITS ENTERED

entered data 82 then RAISE QUERY

PLATELET COUNT [RH_PTC]: __lakhs cells/mm³

FOR BOTH MALES AND FEMALES , 1.50-4.50,

ENTRY SHOULD BE NUMERICAL
VALUES,LIMITED TO 3 DIGITS.DECIMALS
ALLOWED.

RAISE QUERY IF ALPHABET INPUT
ENTERED.

RAISE QUERY IF LESS THAN 2 DIGITS OR
MORE THAN 3 DIGITS ENTERED

Entered data 1.40 then RAISE QUERY

RED BLOOD CELL (RH_RBC) COUNT:

__million cells/microlitre

FOR BOTH MALES AND FEMALES , 4-5

ENTRY SHOULD BE NUMERICAL
VALUES,LIMITED TO 3 DIGITS.DECIMALS
ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF LESS THAN 2 DIGITS OR
MORE THAN 3 DIGITS ENTERED

ENTER DATA 3 ,RAISE QUERY

WHITE BLOOD CELL (RH_WBC) COUNT :

__cells/mm³

FOR BOTH MALES AND FEMALES , 4000-10000

ENTRY SHOULD BE NUMERICAL
VALUES,LIMITED TO 5 DIGITS,NOT LESS
THAN 4 DIGITS.DECIMALS ALLOWED.

RAISE QUERY IF ALPHABET INPUT
ENTERED.

RAISE QUERY IF LESS THAN 4 DIGITS OR
MORE THAN 5 DIGITS ENTERED

RAISE QUERY IF VALUE ENTERED 3999

NEUTROPHILS (RH_NPL):__%

FOR BOTH MALES AND FEMALES , 40-70

ENTRY SHOULD BE NUMERICAL VALUES,
LIMITED TO 3 DIGITS.DECIMALS
ALLOWED.

RAISE QUERY IF ALPHABET INPUT
ENTERED.

RAISE QUERY IF LESS THAN 2 DIGITS OR
MORE THAN 3 DIGITS ENTERED

RAISE QUERY IF VALUE ENTERED 39

LYMPHOCYTES [RH_LMP] :__%

FOR BOTH MALES AND FEMALES , 20-40

ENTRY SHOULD BE NUMERICAL
VALUES,LIMITED TO 3
DIGITS.DECIMALS ALLOWED.

RAISE QUERY IF ALPHABET INPUT
ENTERED.

RAISE QUERY IF LESS THAN 2
DIGITS OR MORE THAN 3 DIGITS
ENTERED

RAISE QUERY IF VALUE ENTERED
19

MONOCYTES [RH_MNC]:__%

FOR BOTH MALES AND FEMALES , 2-10

ENTRY SHOULD BE NUMERICAL
VALUES,LIMITED TO 2
DIGITS.DECIMALS ALLOWED.

RAISE QUERY IF ALPHABET INPUT
ENTERED.

RAISE QUERY IF MORE THAN 2
DIGITS ENTERED

RAISE QUERY IF VALUE ENTERED
11

EOSINOPHILS [RH_ESP] :__%

FOR BOTH MALES AND FEMALES , 0-6

ENTRY SHOULD BE NUMERICAL
VALUES,LIMITED TO 2
DIGIT.DECIMALS ALLOWED.

RAISE QUERY IF ALPHABET INPUT
ENTERED.

RAISE QUERY IF VALUE ENTERED 7

BASOPHILS [RH_BSL]:__%

FOR BOTH MALES AND FEMALES , 0-1

ENTRY SHOULD BE NUMERICAL
VALUES,LIMITED TO 2 DIGIT.
DECIMALS ALLOWED. FOR
EXAMPLE,0.5

RAISE QUERY IF ALPHABET INPUT
ENTERED.

CRF ANNOTATION FOR ROUTINE HEMATOLOGY FORM[RH]

FIELD NAME	FIELD LABEL
RH_HCT	HEMATOCRIT
RH_HB	HAEMOGLOBIN
RH_MCH	MEAN CELLULAR HEMAGLOBIN
RH_MCHC	MEAN CELLULAR HEMOGLOBIN CONCENTRATION
RH_MCV	MEAN CELLULAR VOLUME
RH_PTC	PLATELET COUNT
RH_RBC	RED BLOOD CELL COUNT
RH_WBC	WHITE BLOOD CELL COUNT
RH_NPL	NEUTROPHIL
RH_LMP	LYMPHOCYTE
RH_MNC	MONOCYTE
RH_ESP	EOSINOPHIL
RH_BSL	BASOPHIL

ROUTINE CLINICAL CHEMISTRY [RCC] FORM

ALANINE TRANSAMINASE (RCC_ALT): __ U/L

FOR BOTH MALES AND FEMALES, VALUE LESS THAN 35

ENTRY SHOULD BE A NUMERICAL
VALUE LIMITED TO 3
DIGITS.DECIMAL ALLOWED.

RAISE QUERY IF ALPHABET INPUT
ENTERED

RAISE QUERY IF DATA ENTER 36

ALBUMIN [RCC_ALB]:__ g/Dl

FOR BOTH MALES AND FEMALES, VALUE 3.5-5.2

ENTRY SHOULD BE A NUMERICAL
VALUE LIMITED TO 2 DIGITS.DECIMAL
ALLOWED.

RAISE QUERY IF ALPHABET INPUT
ENTERED

RAISE QUERY IF MORE THAN 2
DIGITS

Raise query if DATA ENTER 3.4

Alkaline phosphatase (RCC_AP): __ U/L

FOR BOTH MALES AND FEMALES, VALUE 33-98

ENTRY SHOULD BE A NUMERICAL
VALUE LIMITED TO 3 DIGITS.DECIMAL
ALLOWED.

RAISE QUERY IF ALPHABET INPUT
ENTERED

RAISE QUERY IF MORE THAN 3 DIGITS

Raise query if DATA ENTER 32

Aspartate transaminase (RCC_AST): __ U/L

FOR BOTH MALES AND FEMALES, VALUE LESS THAN 35

ENTRY SHOULD BE A NUMERICAL
VALUE LIMITED TO 3 DIGITS.DECIMAL
ALLOWED.

RAISE QUERY IF ALPHABET INPUT
ENTERED

RAISE QUERY IF MORE THAN 3 DIGITS

Raise query if DATA ENTERED 36

Bicarbonate [RCC_BCT]: __mmol/L

FOR BOTH MALES AND FEMALES, VALUE 22-29

ENTRY SHOULD BE A NUMERICAL
VALUE LIMITED TO 3 DIGITS.DECIMAL
ALLOWED.

RAISE QUERY IF ALPHABET INPUT
ENTERED

RAISE QUERY IF MORE THAN 3 DIGITS

Raise query if DATA ENTERED 21

Bilirubin – total [RCC_BRT]: __mg/dL

FOR BOTH MALES AND FEMALES, VALUE 0.3-1.2

ENTRY SHOULD BE A NUMERICAL
VALUE LIMITED TO 3 DIGITS.DECIMAL
ALLOWED.

RAISE QUERY IF ALPHABET INPUT
ENTERED

RAISE QUERY IF MORE THAN 3 DIGITS

Raise query if data entered 0.2.

Bilirubin – direct [RCC_BRD] __mg/Dl

FOR BOTH MALES AND FEMALES, VALUE LESS THAN 0.2

***GO TO RCC_BRT , CHECK VALUE ,IF VALUE
IS OUTSIDE REFERENCE RANGE THEN
GENERATE THIS FIELD.***

SHOULD BE NUMERICAL VALUE LIMITED
TO 3 DIGITS.DECIMAL ALLOWED.

RAISE QUERY IF ALPHABET INPUT
ENTERED.

RAISE QUERY IF MORE THAN 3 DIGITS
ENTERED.

RAISE QUERY. IF data entered 0.3

Calcium [RCC_CAL] : __mg/dL

FOR BOTH MALES AND FEMALES, VALUE 8.6-10.6

SHOULD BE NUMERICAL VALUE LIMITED
TO 3 DIGITS.DECIMAL ALLOWED.

RAISE QUERY IF ALPHABET INPUT
ENTERED.

RAISE QUERY IF MORE THAN 3 DIGITS
ENTERED.

RAISE QUERY IF data entered 8.5

Chloride [RCC_CHL]: __mmol/L

FOR BOTH MALES AND FEMALES, VALUE 94-110

SHOULD BE NUMERICAL VALUE LIMITED TO 4 DIGITS.DECIMAL ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF MORE THAN 4 DIGITS ENTERED.

RAISE QUERY IF data entered 93

Creatinine [RCC_CRT]: __mg/dL

FOR BOTH MALES AND FEMALES, VALUE 0.6-1.2

SHOULD BE NUMERICAL VALUE LIMITED TO 3 DIGITS.DECIMAL ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF MORE THAN 3 DIGITS ENTERED.

RAISE QUERY IF VALUE entered 0.5

C-Reactive protein (RCC_CRP): __mg/dL

FOR BOTH MALES AND FEMALES, VALUE less than 1

SHOULD BE NUMERICAL VALUE LIMITED TO 3 DIGITS.DECIMAL ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF MORE THAN 3 DIGITS ENTERED

Raise query if value entered 2

Gamma glutamyl transferase (RCC_GGT): __U/L

FOR BOTH MALES AND FEMALES, VALUE 5-40

SHOULD BE NUMERICAL VALUE LIMITED TO 3 DIGITS.DECIMAL ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF MORE THAN 3 DIGITS ENTERED

Raise query if value entered 4

Glucose [RCC_GLU]

FOR BOTH MALES AND FEMALES, VALUE LESS THAN 100

SHOULD BE NUMERICAL VALUE LIMITED TO 4 DIGITS.DECIMAL ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF MORE THAN 4 DIGITS ENTERED

Raise query if value 102

Insulin [RCC_INS]: __μIU/mL

FOR BOTH MALES AND FEMALES, VALUE 2-25

SHOULD BE NUMERICAL VALUE LIMITED TO 3 DIGITS.DECIMAL ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF MORE THAN 3 DIGITS ENTERED

Raise query if value entered 26

HbA1c [RCC_HBC]: __%

FOR BOTH MALES AND FEMALES, VALUE less than 5.7

SHOULD BE NUMERICAL VALUE LIMITED TO 2 DIGITS.DECIMAL ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF MORE THAN 2 DIGITS ENTERED

Raise query if value entered 5.8

Potassium [RCC_PTS]: __mmol/L

FOR BOTH MALES AND FEMALES, VALUE 3.5-5

SHOULD BE NUMERICAL VALUE LIMITED TO 3 DIGITS.DECIMAL ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF MORE THAN 3 DIGITS ENTERED

Raise query if value entered 5.2

Total cholesterol, [RCC_TCL] : __mg/dL

FOR BOTH MALES AND FEMALES, VALUE less than 200

SHOULD BE NUMERICAL VALUE LIMITED TO 4 DIGITS.DECIMAL ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF MORE THAN 4 DIGITS ENTERED

Raise query if value entered 181

LDL cholesterol, [RCC_LDL] __mg/dL

FOR BOTH MALES AND FEMALES, VALUE less than 70

SHOULD BE NUMERICAL VALUE LIMITED TO 3 DIGITS.DECIMAL ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF MORE THAN 3 DIGITS ENTERED

Raise query if value entered 71.

HDL cholesterol [RCC_HDL] __mg/dL

FOR BOTH MALES AND FEMALES, VALUE less than 40

SHOULD BE NUMERICAL VALUE LIMITED TO 3 DIGITS.DECIMAL ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF MORE THAN 3 DIGITS ENTERED

Raise query if value entered 41

Triglycerides [RCC_TGLC] __mg/dL

FOR BOTH MALES AND FEMALES, VALUE less than 150

SHOULD BE NUMERICAL VALUE LIMITED TO 4 DIGITS.DECIMAL ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF MORE THAN 4 DIGITS ENTERED

Raise query if value entered 151

Protein – total [RCC_PRT]: __g/dL

FOR BOTH MALES AND FEMALES, VALUE 6-8.3

SHOULD BE NUMERICAL VALUE LIMITED TO 3 DIGITS.DECIMAL ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF MORE THAN 3 DIGITS ENTERED

Raise query if value entered 8.4

Sodium [RCC_SDM]: __mEq/L

FOR BOTH MALES AND FEMALES, VALUE 135-145

SHOULD BE NUMERICAL VALUE LIMITED TO 4 DIGITS.DECIMAL ALLOWED.

RAISE QUERY IF ALPHABET INPUT ENTERED.

RAISE QUERY IF MORE THAN 4 DIGITS ENTERED

Raise query if value entered 132

CRF ANNOTATION FOR ROUTINE CLINICAL CHEMISTRY FORM [RCC]

FIELD NAME	FIELD LABEL
RCC_ALT	ALANINE TRANSAMINASE
RCC_ALB	ALBUMIN
RCC_AP	ALKALINE PHOSPHATASE
RCC_AST	ASPARTATE TRANSAMINASE
RCC_BCT	BICARBONATE
RCC_BRT	BILIRUBIN-TOTAL
RCC_BRD	BILIRUBIN-DIRECT
RCC_CAL	CALCIUM
RCC_CHL	CHLORIDE
RCC_CRT	CREATININE
RCC_CRP	C REACTIVE PROTEIN
RCC_GGT	GAMMA GLUTAMYL TRANSFERASE
RCC_GLU	GLUCOSE
RCC_INS	INSULIN
RCC_HBC	HbA1C
RCC_PST	POTASSIUM
RCC_TCL	TOTAL CHOLESTROL
RCC_LDL	LDL CHOLESTROL
RCC_HDL	HDL CHOLESTROL
RCC_TGLC	TRIGLYCERIDES
RCC_PRT	PROTEIN-TOTAL
RCC_SDM	SODIUM